

DAILY DRILLING REPORT SET

SYN-Assam-Duliajan-002 (W002) — SYN-Assam-Duliajan

1. REPORT SET IDENTIFICATION

Well ID	W002
Well Name	SYN-Assam-Duliajan-002
Field / Cluster	SYN-Assam-Duliajan
Area	Synthetic Assam Field Cluster (Duliajan-type operating area)
Latitude / Longitude	27.389232 N / 95.288712 E
Rig	SYN-RIG-N1600-02
Spud Date	12-Feb-2016
Final Report Date	23-Feb-2016
Daily Reports in Set	12 (Day 01 to Day 12)
Total Depth Reached	3,450 m
Primary Formation	Sandstone
Events Reported in Set	7

2. DEPTH PROGRESSION SUMMARY

Day	Date	From (m)	To (m)	Drilled (m)	Formation	Events
01	12-Feb-2016	0	480	480	Claystone	Mud Loss
02	13-Feb-2016	480	955	475	Claystone / Shale	—
03	14-Feb-2016	955	1,235	280	Shale	Wellbore Instability, Stuck Pipe
04	15-Feb-2016	1,235	1,485	250	Shale	Cementing Issue
05	16-Feb-2016	1,485	1,795	310	Shale	Stuck Pipe
06	17-Feb-2016	1,795	2,115	320	Shale / Sandstone	—
07	18-Feb-2016	2,115	2,410	295	Sandstone	—
08	19-Feb-2016	2,410	2,685	275	Sandstone	—
09	20-Feb-2016	2,685	2,825	140	Sandstone	Mud Loss
10	21-Feb-2016	2,825	3,065	240	Sandstone	—
11	22-Feb-2016	3,065	3,290	225	Sandstone / Limestone	—
12	23-Feb-2016	3,290	3,450	160	Limestone	Kick

PROTOTYPE TRAINING RECORD. Synthetic demonstration document prepared for the eRTMAC-NWIS prototype (SIH PS 26121). Depths, dates, parameters and events are generated from the prototype datasets and are not operational records of any company.

DAILY DRILLING REPORT — DAY 01						W002 · 12-Feb-2016														
Well ID: W002 Well Name: SYN-Assam-Duliajan-002 Report Date: 2016-02-12 Day Number: 1 Start Depth: 0 m End Depth: 480 m Drilled Interval: 480 m Formation: Claystone Mud Weight: 9.34 ppg ROP: 33.1 m/hr RPM: 142 WOB: 6.8 klbf Events: Mud Loss at 476 m (Low) NPT: 2.2 hours																				
Operations performed. - Rig up completed and well spudded at 12-Feb-2016. Drilled 17-1/2" hole from surface to 480 m. - Made connections and circulated bottoms-up at intervals; shakers monitored. - Mud Loss handled at 476 m — 2.2 hr non-productive time. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.																				
Geological observations. Cuttings throughout the interval are claystone, consistent with the 0–785 m unit logged in the offset wells of this cluster.																				
Mud / circulation observations. Returns reduced during the tour; losses treated with LCM and mud weight trimmed. Loss volumes recorded on the daily loss sheet. Mud weight held at 9.34 ppg after treatment.																				
Equipment / operational observations. BOP drill carried out with the crew; response time satisfactory.																				
Events reported this day. <table><tr><th>Event ID</th><th>Depth (m)</th><th>Formation</th><th>Type</th><th>Severity</th><th>Hr</th><th>Operational Impact</th></tr><tr><td>E0021</td><td>476</td><td>Claystone</td><td>Mud Loss</td><td>Low</td><td>2.2</td><td>Minor non-productive time; operations resumed after routine parameter adjustment.</td></tr></table>							Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact	E0021	476	Claystone	Mud Loss	Low	2.2	Minor non-productive time; operations resumed after routine parameter adjustment.
Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact														
E0021	476	Claystone	Mud Loss	Low	2.2	Minor non-productive time; operations resumed after routine parameter adjustment.														
End-of-day status. Depth 480 m in Claystone. Drilling ahead. Non-productive time this tour 2.2 hr.																				

DAILY DRILLING REPORT — DAY 02						W002 · 13-Feb-2016
Well ID: W002 Well Name: SYN-Assam-Duliajan-002 Report Date: 2016-02-13 Day Number: 2 Start Depth: 480 m End Depth: 955 m Drilled Interval: 475 m Formation: Claystone / Shale Mud Weight: 10.07 ppg ROP: 25.3 m/hr RPM: 137 WOB: 11.5 klbf Events: None NPT: 0.0 hours						
Operations performed. - Drilled 475 m of section from 480 m to 955 m (16.5 hr on bottom). - Deviation survey taken; hole angle within tolerance. - Pulled out of hole, ran 13-3/8" casing to 785 m and cemented as programmed (Lead 12.6 ppg / Tail 15.8 ppg, cement to surface). Waited on cement and pressure tested.						
Geological observations. Formation change crossed during the tour: Claystone to 785 m, Shale to 1,880 m. Cuttings confirm the tops used for correlation with adjacent wells in the cluster.						
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 10.07 ppg, properties held within programme; no losses or gains recorded.						
Equipment / operational observations. All rig equipment functioning normally; no downtime recorded.						
Events reported this day. Prototype Training Record · Synthetic demonstration data None. Operations proceeded within the planned parameter envelope.						

DAILY DRILLING REPORT — DAY 03						W002 · 14-Feb-2016
Well ID: W002 Well Name: SYN-Assam-Duliajan-002 Report Date: 2016-02-14 Day Number: 3 Start Depth: 955 m End Depth: 1235 m Drilled Interval: 280 m Formation: Shale Mud Weight: 10.65 ppg ROP: 17.1 m/hr RPM: 127 WOB: 15.7 klbf Events: Wellbore Instability at 1098 m (High); Stuck Pipe at 1128 m (Medium) NPT: 37.6 hours						
Operations performed. - Drilled 280 m of Shale from 955 m to 1,235 m (2.0 hr on bottom). - Made connections and circulated bottoms-up at intervals; shakers monitored. - Wellbore Instability handled at 1,098 m — 26.1 hr non-productive time. Mud inhibition and rheology increased, sweeps circulated to clear cavings and controlled back-reaming carried out; ROP limited through the unstable interval. - Stuck Pipe handled at 1,128 m — 11.5 hr non-productive time. String worked and jarred within the allowable overpull; lubricant pill spotted and hole cleaning improved with high-viscosity sweeps and increased rotation before back-reaming.						
Geological observations. Cuttings throughout the interval are shale, consistent with the 785–1,880 m unit logged in the offset wells of this cluster.						
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 10.65 ppg, properties held within programme; no losses or gains recorded.						
Equipment / operational observations. Overpull recorded on the string during the tour; jars worked and torque monitored. Top drive and drawworks otherwise functioning normally.						
Events reported this day.						
Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact
E0026	1,098	Shale	Wellbore Instability	High	26.1	Significant non-productive time; remedial operations and additional materials required.
E0022	1,128	Shale	Stuck Pipe	Medium	11.5	Partial non-productive time; drilling suspended while the well was treated and conditioned.
End-of-day status. Depth 1,235 m in Shale. Drilling ahead. Non-productive time this tour 37.6 hr.						

DAILY DRILLING REPORT — DAY 04						W002 · 15-Feb-2016
Well ID: W002 Well Name: SYN-Assam-Duliajan-002 Report Date: 2016-02-15 Day Number: 4 Start Depth: 1235 m End Depth: 1485 m Drilled Interval: 250 m Formation: Shale Mud Weight: 11.03 ppg ROP: 17.1 m/hr RPM: 125 WOB: 17.4 klbf Events: Cementing Issue at 1401 m (High) NPT: 20.5 hours						
Operations performed. - Drilled 250 m of Shale from 1,235 m to 1,485 m (2.0 hr on bottom). - Deviation survey taken; hole angle within tolerance. - Cementing Issue handled at 1,401 m — 20.5 hr non-productive time. Job evaluated against the pump/returns record; remedial squeeze / top-up cementing carried out and a cement bond log (CBL/VDL) run to confirm coverage.						
Geological observations. Cuttings throughout the interval are shale, consistent with the 785–1,880 m unit logged in the offset wells of this cluster.						
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 11.03 ppg, properties held within programme; no losses or gains recorded.						
Equipment / operational observations.						

DAILY DRILLING REPORT — DAY 05						W002 · 16-Feb-2016
Well ID: W002 Well Name: SYN-Assam-Duliajan-002 Report Date: 2016-02-16 Day Number: 5 Start Depth: 1485 m End Depth: 1795 m Drilled Interval: 310 m Formation: Shale Mud Weight: 11.43 ppg ROP: 15.7 m/hr RPM: 121 WOB: 19.4 klbf Events: Stuck Pipe at 1506 m (Low) NPT: 5.5 hours						
Operations performed. - Drilled 310 m of Shale from 1,485 m to 1,795 m (17.0 hr on bottom). - Slipped and cut drilling line; routine BOP function test carried out. - Stuck Pipe handled at 1,506 m — 5.5 hr non-productive time. String worked and jarred within the allowable overpull; lubricant pill spotted and hole cleaning improved with high-viscosity sweeps and increased rotation before back-reaming.						
Geological observations. Cuttings throughout the interval are shale, consistent with the 785–1,880 m unit logged in the offset wells of this cluster.						
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 11.43 ppg, properties held within programme; no losses or gains recorded.						
Equipment / operational observations. Overpull recorded on the string during the tour; jars worked and torque monitored. Top drive and drawworks otherwise functioning normally.						
Events reported this day.						
Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact
E0024	1,506	Shale	Stuck Pipe	Low	5.5	Minor non-productive time; operations resumed after routine parameter adjustment.
End-of-day status. Depth 1,795 m in Shale. Drilling ahead. Non-productive time this tour 5.5 hr.						

DAILY DRILLING REPORT — DAY 06						W002 · 17-Feb-2016
Well ID: W002 Well Name: SYN-Assam-Duliajan-002 Report Date: 2016-02-17 Day Number: 6 Start Depth: 1795 m End Depth: 2115 m Drilled Interval: 320 m Formation: Shale / Sandstone Mud Weight: 11.74 ppg ROP: 16.1 m/hr RPM: 119 WOB: 21.2 klbf Events: None NPT: 0.0 hours						
Operations performed. - Drilled 320 m of section from 1,795 m to 2,115 m (16.5 hr on bottom). - Made connections and circulated bottoms-up at intervals; shakers monitored. - Pulled out of hole, ran 9-5/8" casing to 1,880 m and cemented as programmed (Lead 12.9 ppg / Tail 15.8 ppg, TOC 670 m). Waited on cement and pressure tested.						
Geological observations. Formation change crossed during the tour: Shale to 1,880 m, Sandstone to 3,083 m. Cuttings confirm the tops used for correlation with adjacent wells in the cluster.						
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 11.74 ppg, properties held within programme; no losses or gains recorded.						
Equipment / operational observations. Solids-control equipment cleaned; shaker screens changed. No downtime.						
Events reported this day. None. Operations proceeded within the planned parameter envelope.						
End-of-day status.						

DAILY DRILLING REPORT — DAY 07

W002 · 18-Feb-2016

Well ID: W002
Well Name: SYN-Assam-Duliajan-002
Report Date: 2016-02-18
Day Number: 7
Start Depth: 2115 m
End Depth: 2410 m
Drilled Interval: 295 m
Formation: Sandstone
Mud Weight: 12.08 ppg
ROP: 16.1 m/hr
RPM: 117
WOB: 23.2 klbf
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 295 m of Sandstone from 2,115 m to 2,410 m (22.5 hr on bottom).
- Slipped and cut drilling line; routine BOP function test carried out.

Geological observations.

Cuttings throughout the interval are sandstone, consistent with the 1,880–3,083 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 12.08 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

Mud pump No. 2 packing changed during routine circulation; no lost time.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 2,410 m in Sandstone. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 08

W002 · 19-Feb-2016

Well ID: W002
Well Name: SYN-Assam-Duliajan-002
Report Date: 2016-02-19
Day Number: 8
Start Depth: 2410 m
End Depth: 2685 m
Drilled Interval: 275 m
Formation: Sandstone
Mud Weight: 12.54 ppg
ROP: 15.9 m/hr
RPM: 111
WOB: 26.0 klbf
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 275 m of Sandstone from 2,410 m to 2,685 m (22.5 hr on bottom).
- Circulated and conditioned mud; hi-vis sweep circulated bottoms-up.

Geological observations.

Cuttings throughout the interval are sandstone, consistent with the 1,880–3,083 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 12.54 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

Mud pump No. 2 packing changed during routine circulation; no lost time.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 2,685 m in Sandstone. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 09						W002 · 20-Feb-2016
Well ID: W002 Well Name: SYN-Assam-Duliajan-002 Report Date: 2016-02-20 Day Number: 9 Start Depth: 2685 m End Depth: 2825 m Drilled Interval: 140 m Formation: Sandstone Mud Weight: 12.76 ppg ROP: 16.3 m/hr RPM: 109 WOB: 26.4 klbf Events: Mud Loss at 2800 m (Critical) NPT: 18.7 hours						
Operations performed. - Drilled 140 m of Sandstone from 2,685 m to 2,825 m (3.8 hr on bottom). - Circulated and conditioned mud; hi-vis sweep circulated bottoms-up. - Mud Loss handled at 2,800 m — 18.7 hr non-productive time. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.						
Geological observations. Cuttings throughout the interval are sandstone, consistent with the 1,880–3,083 m unit logged in the offset wells of this cluster.						
Mud / circulation observations. Returns reduced during the tour; losses treated with LCM and mud weight trimmed. Loss volumes recorded on the daily loss sheet. Mud weight held at 12.76 ppg after treatment.						
Equipment / operational observations. All rig equipment functioning normally; no downtime recorded.						
Events reported this day.						
Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact
E0002	2,800	Sandstone	Mud Loss	Critical	18.7	Major non-productive time; section objectives affected and specialist intervention required.
End-of-day status. Depth 2,825 m in Sandstone. Drilling ahead. Non-productive time this tour 18.7 hr.						

DAILY DRILLING REPORT — DAY 10						W002 · 21-Feb-2016
Well ID: W002 Well Name: SYN-Assam-Duliajan-002 Report Date: 2016-02-21 Day Number: 10 Start Depth: 2825 m End Depth: 3065 m Drilled Interval: 240 m Formation: Sandstone Mud Weight: 13.05 ppg ROP: 13.8 m/hr RPM: 107 WOB: 28.2 klbf Events: None NPT: 0.0 hours						
Operations performed. - Drilled 240 m of Sandstone from 2,825 m to 3,065 m (22.5 hr on bottom). - Slipped and cut drilling line; routine BOP function test carried out.						
Geological observations. Cuttings throughout the interval are sandstone, consistent with the 1,880–3,083 m unit logged in the offset wells of this cluster.						
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 13.05 ppg, properties held within programme; no losses or gains recorded.						
Equipment / operational observations. Solids-control equipment cleaned; shaker screens changed. No downtime.						
Events reported this day. None. Operations proceeded within the planned parameter envelope.						
End-of-day status Probing Test Log Record. Synthetic demonstration data. Depth 3,065 m in Sandstone. Drilling ahead. No non-productive time this tour.						

DAILY DRILLING REPORT — DAY 11

W002 · 22-Feb-2016

Well ID: W002
Well Name: SYN-Assam-Duliajan-002
Report Date: 2016-02-22
Day Number: 11
Start Depth: 3065 m
End Depth: 3290 m
Drilled Interval: 225 m
Formation: Sandstone / Limestone
Mud Weight: 13.76 ppg
ROP: 8.7 m/hr
RPM: 94
WOB: 33.4 klbf
Events: None
NPT: 0.0 hours

Operations performed.

• Drilled 225 m of section from 3,065 m to 3,290 m (22.5 hr on bottom).

• Made connections and circulated bottoms-up at intervals; shakers monitored.

Geological observations.

Formation change crossed during the tour: Sandstone to 3,083 m, Limestone to 3,450 m. Cuttings confirm the tops used for correlation with adjacent wells in the cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 13.76 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

BOP drill carried out with the crew; response time satisfactory.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 3,290 m in Limestone. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 12

W002 · 23-Feb-2016

Well ID: W002
Well Name: SYN-Assam-Duliajan-002
Report Date: 2016-02-23
Day Number: 12
Start Depth: 3290 m
End Depth: 3450 m
Drilled Interval: 160 m
Formation: Limestone
Mud Weight: 14.01 ppg
ROP: 7.3 m/hr
RPM: 89
WOB: 35.6 klbf
Events: Kick at 3307 m (Medium)
NPT: 9.8 hours

Operations performed.

• Drilled 160 m of Limestone from 3,290 m to 3,450 m (6.7 hr on bottom).

• Circulated and conditioned mud; hi-vis sweep circulated bottoms-up.

• Pulled out of hole, ran 7" casing to 3,450 m and cemented as programmed (Tail 15.8 ppg, TOC 1,730 m). Waited on cement and pressure tested.

• Reached driller's total depth of 3,450 m. Circulated hole clean, logged and secured the well.

• Kick handled at 3,307 m — 9.8 hr non-productive time. Flow check positive; well shut in on the annular preventer, SIDPP/SICP recorded and kill mud circulated using the driller's method; mud weight raised and the well monitored on the trip tank.

Geological observations.

Cuttings throughout the interval are limestone, consistent with the 3,083–3,450 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Pit gain observed and the well shut in for a flow check; kill mud circulated and mud weight raised to 14.01 ppg. Full returns thereafter.

Equipment / operational observations.

Solids-control equipment cleaned; shaker screens changed. No downtime.

Events reported this day.

Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact
E0025	3,307	Limestone	Kick	Medium	9.8	Partial non-productive time; drilling suspended while the well was treated and conditioned.

End-of-day status.

Prototype Training Record - Synthetic demonstration data

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